Chapter 4

Boolean Algebra and Logic Simplification

（布尔代数和逻辑简化）

4-1 BOOLEAN OPERATIONS AND EXPRESSIONS
（布尔运算和表达式）

- Boolean algebra is the mathematics of digital systems.
  (布尔代数是数字系统的数学工具)

- *A* variable is a symbol (usually an italic uppercase letter) used to represent a logic quantity. Any single variable can have a 1 or 0 value.
  （通常用一个斜体大写字母表示一个变量。一个单变量的取值为0或者1）

4-1 BOOLEAN OPERATIONS AND EXPRESSIONS
（布尔运算和表达式）

- The complement is the inverse of a variable and is indicated by a bar over the variable (overbar). The complement of the variable *A* is read as "not *A*" or "*A* bar". Sometimes a prime symbol rather than an overbar is used to denote the complement of a variable.

- 一个变量的反码也称之为其补码，通常用符号上面的一横表示。读作"not *A*" or "*A* bar"。有时候也用一撇代替一横杠，用来表示一个变量的反码/补码

$$A \qquad \overline{A} \qquad A'$$

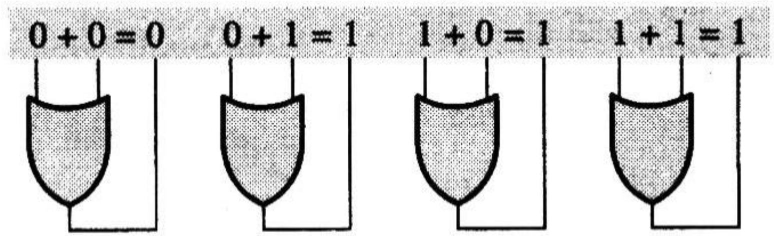
$$A+B, \quad A+B+\overline{C}, \quad \overline{A}+B+C+\overline{D}$$

$$AB, \quad AB\overline{C}, \quad \overline{A}BC\overline{D}$$

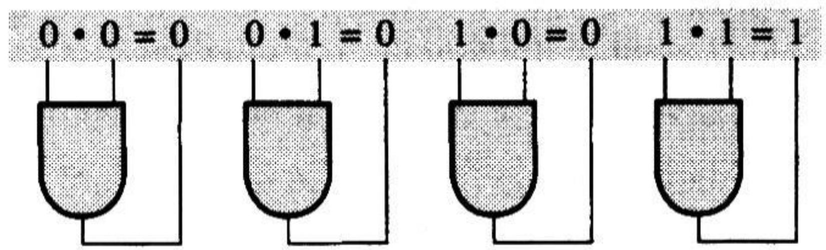
- Literal（因子）: Variable or complement of a variable

Boolean Operations and Expressions

- Boolean addition is equivalent to the OR operation and the basic rules are illustrated with their relation to the OR gates as follows:



- Boolean multiplication is equivalent to the AND operation and the basic rules are illustrated with their relation to the AND gates as follows:



AND (Boolean Multiplication) : $\cdot$

e.g. A AND B = A $\cdot$B = AB

OR (Boolean Addition): +

e.g. A OR B = A+B

Inverter(NOT): $\bar{}$

e.g. $\overline{A}$

sum term (求和项)

sum term: sum of literals. (Literal（因子）：Variable or complement of a variable)

e.g. $A + B,\ A + \overline{B} + \overline{C},\ \overline{A} + B + \overline{C} + D$

Example:

Determine the value of A, B, C, and D which make the sum term $A + \overline{B} + C + \overline{D} = 0$

Solution:

$$A + \overline{B} + C + \overline{D} = 0 \Rightarrow A = 0, \overline{B} = 0, C = 0, \overline{D} = 0$$

$$\Rightarrow A = 0, B = 1, C = 0, D = 1$$

product term（乘积项）

product term: product of literals.

e.g. $AB,\ A\bar{B}C,\ A\bar{B}CD$

Example:

Determine the value of A, B, C, and D which make the product term $A\bar{B}C\bar{D} = 1$

Solution:

$$A\bar{B}C\bar{D} = 1 \Rightarrow A = 1, \bar{B} = 1, C = 1, \bar{D} = 1$$

$$\Rightarrow A = 1, B = 0, C = 1, D = 0$$

4-2 LAWS AND RULES OF BOOLEAN ALGEBRA
布尔代数的定律和法则

	Laws	Boolean Expression
1	Commutative law of addition Commutative law of multiplication (交换律)	$A+B=B+A$ $AB=BA$
2	Associative law of addition Associative law of multiplication (结合律)	$A+(B+C)=(A+B)+C$ $A(BC)=(AB)C$
3	Distributive law (分配率)	$A(B+C)=AB+AC$

Laws and Rules of Boolean Algebra (I)

Rule Number	**Boolean Expression**
1	$A + 0 = A$
2	$A + 1 = 1$
3	$A \cdot 0 = 0$
4	$A \cdot 1 = A$
5	$A + A = A$
6	$A + \bar{A} = 1$

Laws and Rules of Boolean Algebra (II)

Rule Number	**Boolean Expression**
7	$A \cdot A = A$
8	$A \cdot \overline{A} = 0$
9	$\overline{\overline{A}} = A$
10	$A + AB = A$
11	$A + \overline{A}B = A + B$
12	$(A+B)(A+C) = A + BC$
13	$AB + \overline{A}C + BC = AB + \overline{A}C$

Laws and Rules of Boolean Algebra _PROOF (I)

- This law is similar to absorption in that it can be employed to eliminate extra elements from a Boolean expression:

$$A + \overline{A}B = A + B$$

[PROOF]

$$A+\overline{A}B = A(1+B)+\overline{A}B = (A+AB)+\overline{A}B$$

$$= A+(AB+\overline{A}B)$$

$$= A+(A+\overline{A})B$$

$$= A+1\cdot B = A+B$$

Laws and Rules of Boolean Algebra _PROOF (II)

$$(A+B)(A+C)=A+BC$$

[PROOF]

$$\begin{aligned}(A+B)(A+C) &= (A+B)A+(A+B)C \\ &= AA+AB+AC+BC \\ &= A+AB+AC+BC \\ &= A(1+B+C)+BC \\ &= A+BC\end{aligned}$$

Laws and Rules of Boolean Algebra (III)

$$AB + \overline{A}C + BC = AB + \overline{A}C$$

[PROOF]

$$AB + \overline{A}C + BC = AB + \overline{A}C + (A + \overline{A})BC$$

$$= AB + \overline{A}C + ABC + \overline{A}BC$$

$$= (AB + ABC) + (\overline{A}C + \overline{A}CB) = AB + \overline{A}C$$

The key to using this theorem is to find a variable and its complement, note the associated terms, and eliminate the included term (the consensus term), which is composed of the associated terms.

4-3 DEMORGAN'S THEOREMS
狄摩根定理

- ***DeMorgan's first theorem***:
- *The complement of a product of variables is equal to the sum of the complements of the variables.*

$$\overline{XY} = \overline{X} + \overline{Y}$$

(equivalency of the NAND and negative-OR gates)

Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

- ***DeMorgan's second theorem***:
  The complement of a sum of variables is equal to the product of the complements of the variables.

$$\overline{X + Y} = \overline{X}\,\overline{Y}$$

(equivalency of the NOR and negative-AND gates.)

Inputs		Output	
X	Y	$\overline{X+Y}$	$\overline{X}\,\overline{Y}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Apply DeMorgan's theorem to the expressions:

$$\overline{XYZ}, \overline{X + Y + Z}$$

Solution:

$$\overline{XYZ} = \overline{X} + \overline{YZ} = \overline{X} + (\overline{Y} + \overline{Z})$$

$$= \overline{X} + \overline{Y} + \overline{Z}$$

$$\overline{X + Y + Z} = \overline{X} \cdot \overline{Y + Z} = \overline{X}\,\overline{Y}\,\overline{Z}$$

Apply DeMorgan's theorems to each of the following expressions:

(a) $\overline{(A+B+C)D}$ (b) $\overline{ABC+DEF}$

Solution:

(a) Let $A+B+C=X$, $D=Y$:

$\overline{(A+B+C)D} \Rightarrow \overline{XY} = \overline{X} + \overline{Y} = \overline{A+B+C} + \overline{D}$

$\overline{A+B+C} \Rightarrow \overline{A+B+C} = \overline{A}\,\overline{B}\,\overline{C}$

$\overline{(A+B+C)D} \Rightarrow \overline{A}\,\overline{B}\,\overline{C} + \overline{D}$

$$(b) \quad \overline{ABC+DEF} = (\overline{ABC})(\overline{DEF})$$
$$= (\overline{A}+\overline{B}+\overline{C})(\overline{D}+\overline{E}+\overline{F})$$

Example: Apply DeMorgan's theorem to the following expression: $\overline{\overline{A+B\overline{C}}+D(\overline{E+\overline{F}})}$

Solution:

$$\overline{\overline{A+B\overline{C}}+D(\overline{E+\overline{F}})}$$

$$\overline{X+Y}=\overline{X}\,\overline{Y}$$

$$=(\overline{\overline{A+B\overline{C}}})(\overline{D(\overline{E+\overline{F}})})$$

$$\overline{XY}=\overline{X}+\overline{Y}$$

$$=(A+B\overline{C})(\overline{D}+(\overline{\overline{E+\overline{F}}}))$$

$$=(A+B\overline{C})(\overline{D}+E+\overline{F})$$

$$=A\overline{D}+AE+A\overline{F}+B\overline{C}\,\overline{D}+B\overline{C}E+B\overline{C}\,\overline{F}$$

4.4 Logic function Simplification (逻辑函数化简)

Objective:

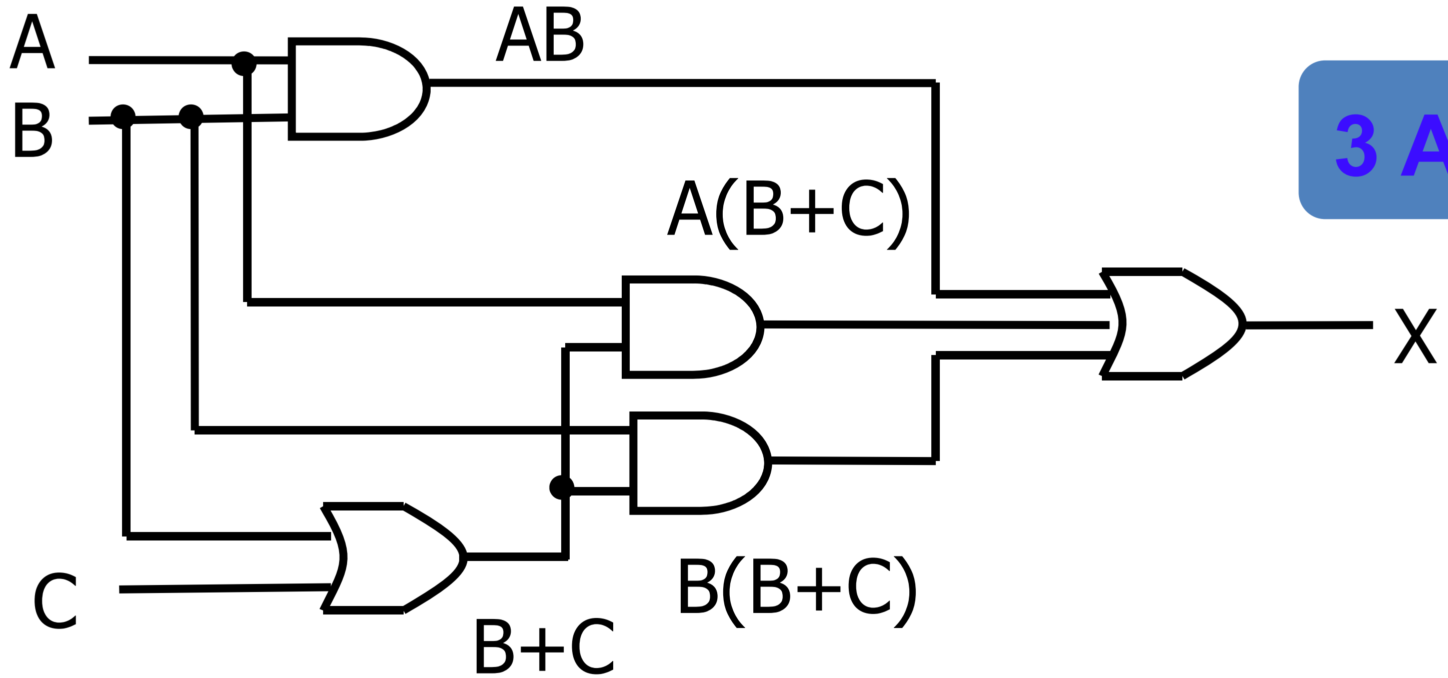
To reduce a particular expression in its ***simplest form*** or ***change its form to a more convenient one*** to implement the expression most efficiently. The approach taken in this section is to use the ***basic laws, rules, and theorems*** of Boolean algebra to simplify an expression.

- Example 4-5 Using Boolean algebra, simplify this expression:

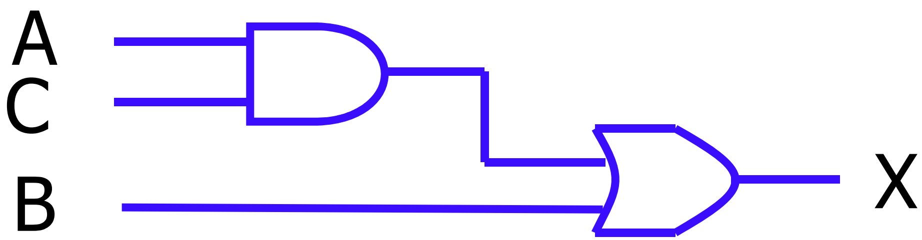
$$X = AB + A(B + C) + B(B + C)$$

- Solution:

$$\begin{aligned} X &= AB + A(B + C) + B(B + C) \\ &= AB + AB + AC + BB + BC \\ &= AB + AB + AC + B + BC \\ &= AB + AC + B + BC \\ &= AB + AC + B \\ &= AC + B \end{aligned}$$



3 AND,2 OR



1 AND,1 OR

Example:

$$(A\overline{B}(C + BD) + \overline{A}\overline{B})C$$

Example:

$$(A\overline{B}(C+BD)+\overline{A}\overline{B})C$$

$$=(A\overline{B}C+A\overline{B}BD+\overline{A}\overline{B})C$$

$$=(A\overline{B}C+A\cdot 0\cdot D+\overline{A}\overline{B})C$$

$$=(A\overline{B}C+\overline{A}\overline{B})C$$

$$=A\overline{B}CC+\overline{A}\overline{B}C$$

$$=A\overline{B}C+\overline{A}\overline{B}C$$

$$=(A+\overline{A})\overline{B}C$$

$$=1\cdot\overline{B}C$$

$$=\overline{B}C$$

4 AND,2 OR

1 AND

Example

$$F = A\overline{\overline{B}CD} + A\overline{B}CD \quad = A$$

$$AB + A\overline{B} = A$$

$$2.\ A + AB = A$$

Example:

$$F = A + \overline{\overline{A} \cdot \overline{BC}}\,(\overline{A} + \overline{\overline{B}\,\overline{C} + D}) + BC$$

$$\overline{X \cdot Y} = \overline{X} + \overline{Y}$$

$$= A + BC + (A + BC)(\overline{A} + \overline{\overline{B}\,\overline{C} + D})$$

$$A + AB = A$$

$$= A + BC$$

Example:

$$F = AC + \overline{A}D + \overline{C}D$$
$$= AC + (\underline{\overline{A} + \overline{C}})D$$

$$\overline{X \cdot Y} = \overline{X} + \overline{Y}$$

$$= AC + \overline{AC}D \qquad A + \overline{A}B = A + B$$
$$= AC + D$$

Example:

$$\boldsymbol{F = \overline{A}B\overline{C} + \overline{A}BC + ABC}$$

$$= \overline{A}B\overline{C} + \underline{\overline{A}BC + ABC}$$

$$= \overline{A}B\overline{C} + BC$$

$$= B(\overline{A}\,\overline{C} + C)$$

$$= B(\overline{A} + C)$$

$$\boldsymbol{= \overline{A}B + BC}$$

Simplification Using Boolean Algebra

- Examples:

$$Y_1 = A\overline{B} + AC + BC$$

$$Y_2 = ABD + A\overline{B}C\overline{D} + A\overline{C}DE + A$$

$$Y_3 = \overline{A}BC + (A + \overline{B})C$$

$$Y_4 = \overline{\overline{A}BC} + \overline{A\overline{B}}$$

$$Y_5 = A\overline{B}(\overline{A}CD + \overline{AD + \overline{B}\,\overline{C}})(\overline{A} + B)$$

$$Y_6 = AC(\overline{C}D + \overline{A}B) + BC\overline{\overline{\overline{B} + AD} + CE}$$

$$Y_7 = \overline{A\overline{B}\,\overline{C}D + A\overline{C}DE + \overline{B}D\overline{E} + A\overline{C}\,\overline{D}E}$$